

A positive answer for Kaufmann's cusp question

FA/Banach automatic math run

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1 Source question

The source paper is P. L. Kaufmann, *Products of Lipschitz-free spaces and applications*, arXiv:1403.6605, Studia Math. 226 (2015), 213–227. After Proposition 5.1, the paper asks:

Consider the set

$$Cusp = \{(x, 0) : x \geq 0\} \cup \{(x, x^2) : x \geq 0\}$$

endowed with the Euclidean metric. Is $\mathcal{F}(Cusp)$ isomorphic to a subspace of L^1 ?

it is the adjoint of an isomorphism from $\mathcal{F}(\cup_{\gamma \in \Gamma} M_\gamma)$ onto $(\sum_{\gamma \in \Gamma} \mathcal{F}(M_\gamma))_{\ell_1}$. It is clear from (5.2) that the distortion of Φ is bounded by C . $\square$

Note that, if we exclude the orthogonality condition, the concatenating operator Φ defined in (5.1) does not need to take values in $Lip_0(M)$. We do not know if it is possible to remove this condition, even for finite unions. In fact, in particular we do not now the answer to the following apparently simple question:

Problem. Consider the set $Cusp := \{(x, 0) : x \geq 0\} \cup \{(x, x^2) : x \geq 0\}$ endowed with the euclidean metric. Is $\mathcal{F}(Cusp)$ isomorphic to a subspace of L^1 ?

Note that $Cusp$ is not Lipschitz equivalent to the real line, so we cannot appeal to Proposition 1.1 to have a positive answer. A negative answer would imply that there is no isomorphic copy of $\mathcal{F}(\mathbb{R}^2)$ in L^1 , which as we mentioned is one of the main results from [13].

The mentioned result by Godard can be seen as a particular case of Proposition 5.1:

Proposition 5.2 (Godard [5], Proposition 5.1). *Let Γ be a set with a distinguished point 0,*

2 Proof intuition

The obstruction in the source question is the tangency at the origin: the two branches are not orthogonal there, so Kaufmann's direct union formula does not apply. The tangency can nevertheless be organized scale by scale.

Place short rungs between the two branches at the points $1/n$ near the origin and at the integers far away. The Euclidean cusp is bi-Lipschitz equivalent to the two rails with the induced metric from this weighted ladder graph. The rung spacing is chosen so that near the origin the horizontal spacing is comparable with the cusp thickness x^2 , while at infinity integer spacing is already fine because the cusp thickness is large.

The full ladder is a metric graph. Its free space is an L^1 -edge space modulo the graph's cycle space. Choosing the small upper edges and the large vertical rungs as chords gives fundamental

cycles whose L^1 -lengths are uniformly controlled by their chord lengths. Therefore the cycle space is complemented in the edge L^1 space. The quotient, hence the ladder free space, is isomorphic to a subspace of L^1 . Finally, Lipschitz functions on the two rails extend linearly over the rungs, so the rail free space is complemented in the ladder free space.

3 The result

Theorem 1. *Let*

$$C = \{(x, 0) : x \geq 0\} \cup \{(x, x^2) : x \geq 0\}$$

with the metric inherited from the Euclidean plane. Then $\mathcal{F}(C)$ is isomorphic to a subspace of an L^1 -space.

Proof. Write

$$a(t) = (t, 0), \quad b(t) = (t, t^2) \quad (t \geq 0).$$

Let

$$P = \{1/n : n \in \mathbb{N}\} \cup \mathbb{N}.$$

We build a metric graph G . Its two rails are copies of the two cusp branches. For each interval

$$[1/(n+1), 1/n] \quad (n \geq 1), \quad [m, m+1] \quad (m \geq 1),$$

put one lower rail edge from $a(p)$ to $a(q)$ and one upper rail edge from $b(p)$ to $b(q)$, each with its Euclidean chord length. For each $p \in P$ add a vertical edge V_p from $a(p)$ to $b(p)$ of length p^2 . Let $R \subset G$ denote the union of the two rails, with the metric induced from the graph G .

Lemma 1. *The identity map from the Euclidean cusp C onto R is bi-Lipschitz.*

Proof. Every edge of G is realized as a Euclidean segment joining the same endpoints in the plane. Therefore every graph path between two rail points has length at least the Euclidean distance between those points, so $d_{\text{Euc}} \leq d_G$ on R .

For points on the lower branch, the graph distance is at most the lower rail length, which is exactly the Euclidean distance. For points $b(s), b(t)$ on the upper branch, the graph distance is at most the polygonal length along the parabola. Since

$$\int_s^t (1 + 4u^2)^{1/2} du \leq 2|s - t|(1 + (s + t)^2)^{1/2} = 2|b(s) - b(t)|,$$

the upper-branch graph distance is bounded by twice the Euclidean distance.

It remains to compare cross distances. Fix $s, t \geq 0$. If $t = 0$ there is nothing to prove. Choose $p \in P$ with

$$|p - t| \leq 2t^2 \quad (0 < t \leq 1), \quad |p - t| \leq 1 \quad (t \geq 1).$$

Such a point exists by the definitions of P . Also $p \leq 3t$ for $0 < t \leq 1$ and $p \leq 2t$ for $t \geq 1$. The graph path from $a(s)$ to $b(t)$ going along the lower rail to $a(p)$, then across V_p , then along the upper rail to $b(t)$, has length at most

$$|s - p| + p^2 + 2|p - t|(1 + (p + t)^2)^{1/2} \leq K(|s - t| + t^2)$$

for a universal constant K . Since

$$|a(s) - b(t)| = ((s - t)^2 + t^4)^{1/2}$$

and $|s-t|+t^2 \leq 2((s-t)^2+t^4)^{1/2}$, this gives $d_G(a(s), b(t)) \leq K'|a(s)-b(t)|$. The other cross case is symmetric after relabeling the endpoints. Thus d_G and the Euclidean cusp metric are bi-Lipschitz equivalent on R . $\square$

Consequently $\mathcal{F}(C)$ is isomorphic to $\mathcal{F}(R)$. We next show that $\mathcal{F}(R)$ embeds into an L^1 -space.

First, $\mathcal{F}(R)$ is complemented in $\mathcal{F}(G)$. Indeed, if $f \in \text{Lip}_0(R)$, define $Ef = f$ on R and extend Ef linearly on every vertical edge V_p . On every rail edge the restriction is already Lipschitz, and on V_p the endpoint oscillation is at most $\|f\|_{\text{Lip}} p^2$. Thus $E : \text{Lip}_0(R) \rightarrow \text{Lip}_0(G)$ is a norm-one pointwise-continuous linear extension operator. Its preadjoint is a norm-one projection from $\mathcal{F}(G)$ onto the canonical copy of $\mathcal{F}(R)$.

It remains to see that $\mathcal{F}(G)$ is isomorphic to a subspace of L^1 . Orient all edges of G , and let $L^1(E)$ be the L^1 -sum of the edge intervals with respect to arclength. For a metric graph, the edge derivative identifies $\text{Lip}_0(G)$ weak-star continuously with a subspace of $L^\infty(E)$, and the preadjoint identifies

$$\mathcal{F}(G) \simeq L^1(E)/Z,$$

where Z is the closed cycle space of the graph. This is the standard metric-graph form of the Arens–Eells representation.

We show that Z is complemented in $L^1(E)$. Let T be the spanning tree consisting of:

- all lower rail edges;
- all vertical edges $V_{1/n}$, $n \geq 1$;
- all upper rail edges over the intervals $[m, m+1]$, $m \geq 1$.

The chords $E \setminus T$ are exactly the small upper rail edges over $[1/(n+1), 1/n]$ and the large vertical edges V_m , $m \geq 2$.

For a small upper chord $e_n = [b(1/(n+1)), b(1/n)]$, its fundamental cycle consists of e_n , the two adjacent vertical rungs, and the lower edge $[a(1/(n+1)), a(1/n)]$. Since

$$1/n - 1/(n+1) \asymp 1/n^2$$

and the upper chord has comparable length, the L^1 -length of this fundamental cycle is bounded by a universal multiple of $|e_n|$.

For a large vertical chord V_m , $m \geq 2$, its fundamental cycle consists of V_m , the lower rail path from $a(1)$ to $a(m)$, the rung V_1 , and the upper rail path from $b(1)$ to $b(m)$. The lower path has length $m-1$, the upper path is bounded by the parabolic arclength from 1 to m , which is $O(m^2)$, and $|V_m| = m^2$. Hence this fundamental cycle also has L^1 -length at most a universal multiple of its chord length.

Let c_e denote the signed indicator of the fundamental cycle of a chord e , oriented so that $c_e = 1$ on e . Define

$$Qg = \sum_{e \in E \setminus T} \left(\frac{1}{|e|} \int_e g \, ds \right) c_e, \quad g \in L^1(E).$$

The chord supports are disjoint, and the preceding two estimates give

$$\|Qg\|_1 \leq C \sum_{e \in E \setminus T} \int_e |g| \, ds \leq C \|g\|_1.$$

Moreover $Q(c_e) = c_e$ for every fundamental cycle, and the usual fundamental-cycle basis argument gives $\text{range } Q = Z$. Thus Z is complemented in $L^1(E)$. Therefore

$$L^1(E)/Z \simeq (I - Q)L^1(E),$$

which is a subspace of $L^1(E)$. Hence $\mathcal{F}(G)$ is isomorphic to a subspace of L^1 , and so is its complemented subspace $\mathcal{F}(R)$. Since $\mathcal{F}(C) \simeq \mathcal{F}(R)$, the theorem follows. $\square$

4 Verification and novelty notes

The proof is qualitative; no optimized distortion constant is claimed. The only outside structural input is the standard representation of a metric-graph free space as the edge L^1 space modulo the closed cycle space. The packet includes the source PDF and a crop of the problem statement.

Bounded novelty check performed before promotion:

- searched the run registry, solution index, attempt index, and proof-gap index for arXiv ‘1403.6605’, the title, Lipschitz-free product keywords, and cusp/ L^1 phrases;
- searched local parsed sources for later citations of Kaufmann’s paper and for the exact cusp phrase;
- searched the web for exact phrases including “F(Cusp)” “ L^1 ”, “1403.6605” “Cusp” “Lipschitz-free”, and “Products of Lipschitz-free spaces” “Cusp”.

No direct later answer to the cusp question was found in that bounded search.

5 References

References

- [1] A. Godard, *Tree metrics and their Lipschitz-free spaces*, Proc. Amer. Math. Soc. 138 (2010), 4311–4320.
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- [3] P. L. Kaufmann, *Products of Lipschitz-free spaces and applications*, Studia Math. 226 (2015), 213–227, arXiv:1403.6605.
- [4] N. Weaver, *Lipschitz algebras*, World Scientific, 1999.